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Green extraction of active compounds in marine microalgae

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Abstract
Marine microalgae serve as photosynthetic cellular factories for producing active natural products. Currently, compounds such as polysaccharides, proteins, lipids, pigments, and polyphenols extracted from marine microalgae exhibit anti-inflammatory, antimicrobial, and antitumor activities. Components such as pigments and unsaturated fatty acids in marine microalgae possess antioxidant properties and contribute to meeting the nutritional needs of the human body. Extracting these active compounds from marine microalgae requires pretreatment. Methods such as ball milling, high-pressure homogenization, microwave, ultrasound, ionic liquids, and enzymatic methods can disrupt the cell walls of marine microalgae, thereby enhancing extraction efficiency. Green extraction processes include pressurized liquid extraction, supercritical extraction, ionic liquids, and deep eutectic solvents. This paper identifies the current bottlenecks in these pretreatment and extraction processes and proposes potential solutions. In silico simulation and artificial intelligent methods are helpful to design new green solvents and separation processes. Further development of green extraction processes for marine microalgae will advance the application of blue biotechnology.

Keywords Microalgae, Deep eutectic liquid, Ionic liquid, Extraction, Pretreatment

Overview & scope

The ocean covers about 70% of the Earth's surface and gave rise to the earliest forms of life on Earth. Marine diatoms play a critical role in the carbon and nitrogen cycles of marine ecosystems [1]. The ocean is rich in microbial, plant, and animal resources, which can be widely applied in fields such as food [2], pharmaceuticals [3], nutritional supplements [4], energy [5], and aquaculture [6]. Therefore, blue biotechnology targeting the development of marine biological resources will unlock the immense potential of marine biological resources [7], while protecting marine ecosystems during resource development [8], thereby achieving the sustainable development of marine biological resources.

Microalgae are photosynthetic protists, including prokaryotic organisms such as cyanobacteria and eukaryotic organisms such as green algae, primarily found in freshwater and seawater. Microalgae are potential cellular factory for biological carbon capture and reducing greenhouse gas emissions. The biomass obtained after carbon fixation can be converted into biofuels such as bioethanol, biodiesel, and biomethane. Microalgae contain high-value-added products such as sugars, proteins, unsaturated fatty acids, pigments, and other bioactive compounds. In recent years, the market potential for microalgae-based food supplements has been steadily increasing, with their primary products being high-value algal compounds such as lutein [9], β-carotene [10], astaxanthin, diadinoxanthin [11], vitamins [12], fucoxanthin [13, 14], and health-promoting fatty acids (FAs):

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docosahexaenoic acid (DHA) [15], eicosapentaenoic acid (EPA) [16], chrysolaminarin [17], and polysaccharides: β-glucan. These compounds have the potential antimicrobial, anticancer, antioxidant, anti-inflammatory activities [3, 4]. Taking omega-3 polyunsaturated fatty acids as an example, the global market in 2019 was US$2.3 billion, and it is expected to grow at an annual rate of 7.44% from 2019 to 2027, with omega-3 polyunsaturated fatty acids derived from microalgae achieving an annual growth rate of 11.9% [18]. Marine microalgae occupy the lowest tier of the marine food chain. It is worth noting that some bioactive compounds discovered in marine invertebrates are associated with microalgae. Therefore, studying the synthesis of bioactive products from microalgae holds significant importance.

The development of microalgal bioproducts involves upstream technologies such as strain selection and large-scale cultivation, as well as downstream technologies such as harvesting, separation and life cycle assessment (LCA) [19] (Fig. 1). Throughout the entire technical process, the extraction and purification of some microalgal bioproducts far exceed the production costs of microalgae [20]. Since these processes typically involve high costs or high energy consumption, they lack economic viability for commercial production. Despite their small size, microalgae possess complex cell wall structures, and there is significant variability among different microalgal species [21, 22]. If proper pretreatment is not performed, extraction efficiency is limited [22, 23]. Therefore, how to separate other high-value components (such as proteins, carbohydrates, and pigments) from microalgae for use in feed, chemicals, cosmetics, and food industries to create sufficient value and make the process economically viable is a significant opportunity and challenge. It should be noted that to fully develop microalgal biomass and reduce production costs, the separation of multiple products is particularly crucial, which also imposes greater demands on the separation process. Currently, most natural product extractions rely on traditional organic solvents. These solvents pose issues such as high volatility, flammability, explosiveness, and toxicity, necessitating the development of new green solvents for extracting bioactive compounds from marine microalgae.

This paper reviews the latest advances in natural bioactive compounds from marine microalgae, pretreatment of marine microalgae, and extraction of natural bioactive compounds from marine microalgae (Fig. 1), and analyzes the current bottlenecks and potential solutions, especially for the multi-product separation and artificial intelligence (AI) - assisted solvent screening.

Marine microalgae as bioactive reservoirs

Marine microalgae consist of major biomass components (proteins, carbohydrates, and lipids) and trace natural products (pigments, vitamins, etc.). The activities of the bioactive compounds in microalgae are not only species-specific [24] but also influenced by cultivation modes and conditions, such as outdoor or indoor [24], different cultivation volume (5 L or 80 L photobioreactor, PBR) [25], cold or dark treatment [26], culture mode (autotrophic or heterotrophic) [27], and the different extraction solvents [27]. As shown in Table 1, the marine microalgae currently studied include *Nannochloropsis* sp., *Westiellopsis* sp., *Staurosirella pinnata*, *Rhodomonas* sp., *Halamphora* sp., *Tetraselmis* sp., *Skeletonema costatum*, *Chaetoceros* sp., *Neolyngbya* sp., and *Chaetoceros calcitrans*. Marine microalgae also include *Chlorella* sp. [28] and *Spirulina* sp. [29], among others. Active compounds include lipids

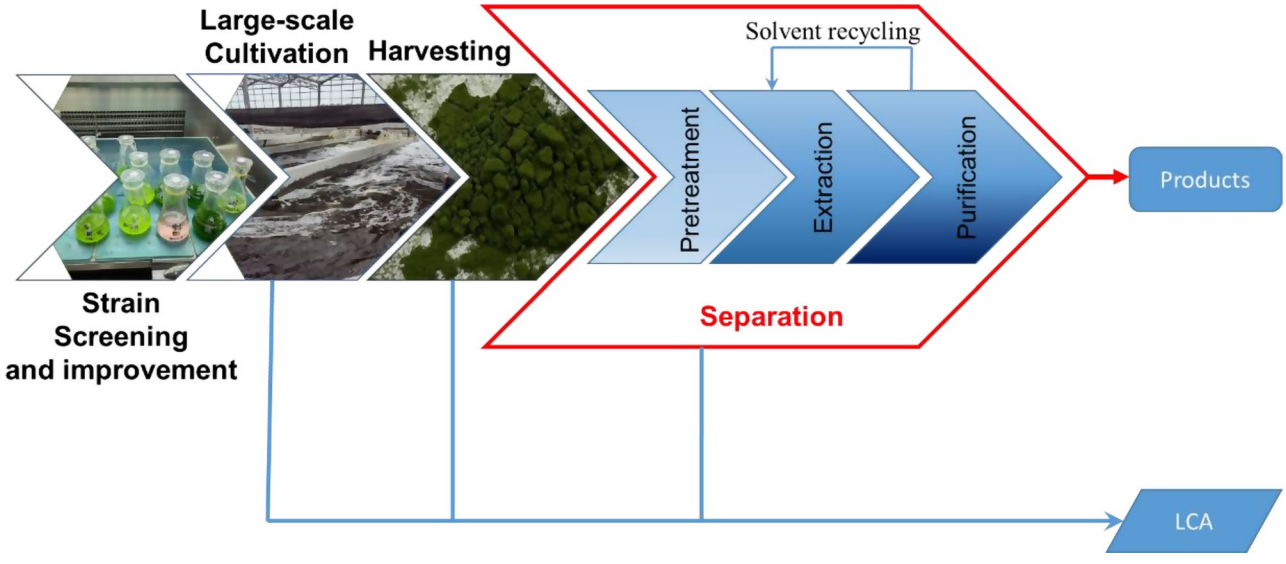

Fig. 1 Upstream processes and downstream processes for marine microalgae development

Table 1 Active compounds and their bioactivities in microalgae

Microalgae	Compounds	Pretreatment	Extraction	Yield or content	Activity type	Activity value	Ref.
Nannochloropsis oceanica	Lipids	Spray-dried	CH_2Cl_2:CH_3OH (2:1, v/v)	26.1% (outdoor); 35.3% (indoor)	Antioxidant	30.4 µmol/g (outdoor); 45.7 µmol/g (indoor);	[24]
Nannochloropsis limnetica	Lipids	Spray-dried	CH_2Cl_2:CH_3OH (2:1, v/v)	23.6% (outdoor); 25.9% (indoor)	Antioxidant	39.5 µmol/g (outdoor); 33.7 µmol/g (indoor);	[24]
Westiellopsis sp.	Phycocyanin	Dried biomass	Dichloromethane: Methanol (2:1)	84.68%	Anticancer	IC_{50}: 120 µg/mL	[95]
Staurosirella pinnata	Lipids	Dried biomass	$CHCl_3$:CH_3OH (2:1 v/v),	-	Pro-thermogenic activity	-	[96]
	Apo-9-fucoxanthinone		CH_3OH:water (20% v/v)	-	Anti-proliferative	-	
Rhodomonas sp.	Proteins	Alcalase and pepsin	Water	5.8–7.5%	Antioxidant	74% at 1 mg/mL	[97]
	Peptides		Water	-	Antioxidant	pepsin (56.5% at 1 mg/mL) and alcalase (47.5% at 1 mg/mL)	
	Peptides	Pepin	Water	-	Anti-Inflammatory	89% of inhibition at 500 µg/mL.	
Halamphora sp. AQ4	Sulfated polysaccharides	Lyophilized biomass	Sulfuric acid	13–50 mg/g	Phagocytosis activity	5 µg/mL	[90]
Tetraselmis chui	Carotenoids	Freeze-dried algal biomass	CH_3OH	8.5 mg/g	Antioxidant	200 µg/mL	[30]
Tetraselmis suecica	Pigments and polyphenols	Lyophilized biomass	50% CH_3OH-water	1–3 mg/g	Antioxidant potential and antiangiogenic Activity	1.38 (5 L PBR) and 1.12 (80 L PBR) mmol Trolox	[25]
T. suecica	peptide	Microalgal pellets	5% acetic acid (ACOH) plus 1x of the halt protease inhibitor cocktail	-	Antimicrobial	-	[98]
T. suecica	Phenols	Dried biomass	Ethanol	3.88 Eq Phloroglucinol (autotrophic); 5.56 Eq Phloroglucinol (heterotrophic)	Antioxidant	3.49 µmol TE/g DW (autotrophic); 6.35 µmol TE/g DW (heterotrophic)	[27]
Skeletonema costatum and Chaetoceros pseudocurvisetus	phenolic compounds	-	Acetone/chloroform	-	Anti-tuberculosis	800 µg/mL	[99]
Neolyngbya (Cyanobacteria)	Crude extracts	Lyophilized biomass	Methanol	27–52%	Antioxidant	10 mg/mL	[100]
Chaetoceros calcitrans (Paulsen) Takano 1968	Fucoxanthin	Freeze-dried samples	Methanol + dichloromethane	49%	Antioxidant activity	4.65 mg TE/g DW	[28]
			Methanol + diethyl ether	43%		2.28 mg TE/g DW	
Chaetoceros calcitrans	Total phenolic compounds	Lyophilized diatom biomass	Methanol, 70% ethanol, acetone, chloroform and hexane	13 mg/g, 19 mg/g, 30 mg/g, 25 mg/g, 10 mg/g	Antioxidant and nitric oxide inhibitory activities	IC_{50} Quercetin = 4.73 µg/mL. IC_{50} Curcumin = 6.10 µg/mL.	[29]

-, none

(Table 2), proteins and polysaccharides (Table 3), pigments (Table 4), and total phenolic compounds. These compounds exhibit antioxidant, antimicrobial, and antitumor activities. The human body cannot synthesize EPA and DHA, which are unsaturated fatty acids, and must obtain them through dietary intake. In the plant kingdom, carotenoid, lutein and other pigments are primarily found in the photosynthetic organelles of higher plants and large algae, including chloroplasts and chromoplasts. Notably, certain fungi, microalgae, and cyanobacteria also possess the ability to biosynthesize these pigments. The value of these compounds in photosynthetic organisms primarily stems from their involvement in light energy conversion processes, where they perform dual regulatory functions. At the cellular level, carotenoids not only maintain membrane structural stability through physical interactions but also possess antioxidant properties that neutralize reactive oxygen species. These unique biological activities confer significant application value in the food industry and healthcare sector, particularly as natural antioxidants that are highly sought after in the market.

Microalgae are considered a natural reservoir of carotenoids, and the types, composition, and cellular localization characteristics of carotenoids in algae are highly similar to those in higher plants. However, due to species-specific metabolic pathways, there are significant differences in the specific composition ratios of carotenoids among different algae. These differences are not only attributed to the genetic characteristics of the species themselves but are also closely related to the lack of standardized extraction processes commonly observed in research. Additionally, environmental parameters (such as light intensity, water salinity, and nutrient concentration) and biological developmental stages (such as growth cycle, harvest season, and water depth) all dynamically regulate carotenoid accumulation within algae. Current research has confirmed that the primary carotenoids in microalgae include β-carotene, lutein, astaxanthin, fucoxanthin, zeaxanthin, and neoxanthin (Table 4).

EPA (C20:5n3), as an important member of the omega-3 series of long-chain polyunsaturated fatty acids [4], plays a central physiological role in cardiovascular regulation, anti-inflammatory effects, and metabolic control, and is widely used in the dietary supplement industry. The omega-3 family primarily consists of alpha-linolenic acid (ALA, C18:3n3), EPA, and DHA (C22:6n3), whose molecular structure and double bond distribution directly influence their biological activity. Traditional fish-based extraction methods face dual challenges: overfishing has disrupted marine ecological balance, and the ocean contamination incident has heightened concerns about the safety of seafood products [18]. This has driven the academic community to actively explore novel biosynthetic pathways, among which EPA biosynthetic technology based on microalgal cell factories has emerged as an international research hotspot due to its sustainable and highly controllable advantages. The primary raw materials for EPA currently come from marine biological resources. The extraction of deep-sea fish faces technical challenges such as poor product stability and noticeable off-flavors. Additionally, bioaccumulation effects result in the presence of heavy metal pollutants (methylmercury, copper) and organic compounds (polychlorinated biphenyls, dioxins) in their bodies, posing risks to human health.

In this context, microalgal biotechnology has demonstrated significant advantages. Microalgal oil not only contains omega-3 polyunsaturated fatty acids comparable to fish oil but also possesses multiple biological characteristics: its lipid components include squalene with anticancer activity and phytosterols that regulate cholesterol, and it is odorless and cholesterol-free. Certain algal species, such as *Nannochloropsis* sp., can efficiently synthesize EPA through photosynthetic autotrophy, with EPA accounting for 20–30% of total lipids within their cells [24]. These species feature short cultivation cycles and high biomass density, enabling large-scale production in photobioreactors, making them an ideal alternative solution to address the resource constraints of traditional fish oil. The EPA synthesis capability of *Nannochloropsis* sp. has attracted widespread attention from both the academic and industrial sectors.

Pretreatment (drying/ball milling/high-pressure homogenization/enzymatic)

Microalgae samples to be processed can be divided into two types: dry algae and wet algae. The process of drying microalgae can also be considered a form of pretreatment. Since fresh microalgae are difficult to preserve, dried microalgae can be stored for longer periods, and more accurate results can be obtained for extraction evaluations. As shown in Table 1, the microalgae samples studied were algae powders obtained through spray drying [24] or freeze drying [30–32]. Oven drying, spray drying, and freeze drying are all energy-intensive processes with limited processing scales. In comparison, sun drying has the lowest cost [33]. For the extraction of marine microalgae, *C. variabilis*, *Scenedesmus regularis*, and *Ankistrodesmus gracilis* typically exhibit higher extraction efficiency in freeze-dried samples compared to spray-dried samples. The C-Phycocyanin extraction yield of freezing and thawing was 73.73 mg/g which was better than those of ultrasonic, maceration and homogenization shown in Table 2 [34]. Some microalgae samples are fresh microalgae, whose extraction conditions generally differ from those of dried algae [35]. Wet algae extraction is typically conducted near microalgae cultivation sites,

Table 2 Extraction of carbohydrates and proteins in marine microalgae

Microalgae	Compounds	Samples	Pretreatment	Extraction	Yield	Purity	Ref.
Isochrysis galbana	Proteins	Lyophilized	-	Chloroform:methanol (2:1) (v/v) mixture→ethanol→acetone→Ionic-Liquid-Based Aqueous Biphasic Systems→Ultrafiltration	100%		[74]
	Glucose rich-polysaccharides		-		71.21%		
Arthospira platensis	C-Phycocyanin	Dry biomass	-	Ultrasonication +Freezing and thawing	109.57 mg/g	0.8	[34]
			-	Ultrasonication + Maceration	99.31 mg/g	0.63	
			-	Ultrasonication	51.51 mg/g	0.62	
				Freezing and thawing	73.73 mg/g	0.66	
				Maceration	55.91 mg/g	0.63	
				Homogenization	52.11 mg/g	0.60	
Nannochloropsis sp.	Protein	Powder	Ultrasonic + choline acetate solution	Methanol and chloroform (1:2 v/v)→ NaOH solution	26.3%	-	[101]
			-	Soxhlet	0.63%	-	
Arthrospira maxima	Phycocyanin	Dried biomass	-	Aqueous extraction →Ultrafiltration	1.174 mg/mL	1.16	[75]
Cyanidium caldarium	Phycocyanin	Algae suspension	Sonication	Ammonium sulphate solution	53 mg/g	0	[41]
			Spark		4 mg/g	0.4	
			Pulsed Electric Fields		0.1	0	
		Purified extracts	Spark Discharges		3 mg/g	1.5	
			Sonication		13 mg/g	1.8	

and the extracted active compounds require prompt processing. Some marine *Spirulina* or *Chlorella* produce extracellular polysaccharides that do not require cell wall disruption for extraction [36]. Both dried biomass and wet biomass were used for the extraction of β-Carotenoid [37]. Using glass beads pretreatment and solvents (ethanol, acetate, methanol and tetrahydrofunan), the carotenoid contents of wet biomass were 2.4–3.0 mg/g DW which were higher than those of dried biomass (1.1–2.3 mg/g DW) [37] (Table 4). It was indicated that drying may lead to the loss of pigment.

Lutein, carotene, and zeaxanthin in marine *Chlorella* can be consumed as food supplements without extraction [28]. In most cases, these active components need to be extracted from marine microalgae. The key bottleneck in the industrialization of microalgae lipids lies in the high-energy-consuming pretreatment steps of cell disruption and lipid extraction. Microalgal cell walls have a complex polysaccharide-protein composite structure, and their high mechanical strength leads to issues such as low energy efficiency and incomplete lipid extraction in traditional disruption methods. Current pretreatment technologies are primarily divided into mechanical and chemical methods: the former includes high-pressure homogenization [38], ultrasonic treatment [39], microwave-assisted [40], and pulsed electric field [41] technologies; the latter includes acid hydrolysis [42, 43], surfactant treatment, and enzymatic hydrolysis [44]. Notably, chemical methods have greater application potential in large-scale production due to their process scalability advantages. The lipid extraction stage typically employs a combination of polar solvents (methanol) and non-polar solvents (hexane). However, the environmental compatibility of solvents remains a pressing issue, and the development of low-toxicity, renewable green solvents has become an important research direction in the biodiesel field.

Ball milling involves adding microspheres to a microalgae solution and using friction and collision between the microspheres to break down the microalgae cell walls. Friction between the microspheres generates heat, so ball milling requires temperature control to prevent the microalgae active products from being inactivated at high temperatures. The optimized lipid extraction conditions for *Schizochytrium* sp. DT3 were bead size 0.4–0.6 μm, 4500 rpm, 4 min when the biomass concentration was 5 g [45]. Optimum conditions achieved for *Tetraselmis suecica* were 1 mm beads, 2000 rpm and the biomass concentration was 100 g/L [46]. Ball milling is also limited by scale and has high energy consumption. High-pressure homogenization can be used for large-scale microalgal cell lysis, but for *Nannochloropsis* sp., even when pressure is increased to 900 bar, the cell lysis rate is only 40% [38]. This also demonstrates that microalgal cell lysis is not straightforward.

Enzyme-catalyzed extraction technology has garnered increasing attention in recent years [44]. It was proved that the lipid yields with enzyme pretreatment were higher than those without enzyme pretreatment for *N. gaditana* [47] (Table 3). Compared to traditional mechanical pretreatment processes, this biocatalytic method demonstrates significant advantages due to its unique biochemical characteristics: first, enzyme preparations can disrupt cell wall structures through specific hydrolysis; Second, its selective catalytic mechanism avoids equipment corrosion while reducing energy consumption; third, the mild aqueous reaction environment effectively prevents the degradation of heat-sensitive active components. Cellulase is typically chosen for cell wall disruption [48]. The combined use of Cellulyve and Feedlyve GMA increased oil extraction efficiency from *Microcystis* by nearly 1.8 times [49]. The use of a crude enzyme mixture of cellulase and amylase doubled protein extraction and increased sugar extraction by 41% in *Oocystis* sp. [50]. By prioritizing the enzymatic decomposition of cell wall matrix components, the resistance to solubilization of target products during subsequent solvent extraction is reduced, while avoiding structural damage to heat-sensitive pigments caused by strong chemical reagents. This biotechnological rate-limiting strategy has demonstrated significant application value in the industrial extraction of carotenoids from microalgae. However, the practical application of this technology still faces challenges, primarily including the need for precise matching of enzymatic reactions with cell wall polysaccharide composition, as well as cost issues in industrial production-enzyme preparations are expensive and require strict maintenance of reaction parameters. Enzyme lytic kit was better than ultrasonication for the protein extractions in marine microalgae (*Tetraselmis* sp. and *Nannochloropsis* sp.) and other four freshwater microalgae [51]. The components of cell walls of different microalgae vary greatly. The enzymatic pretreatment should be well-designed and it is expensive than the mechanical methods.

Solvent systems (traditional/ILs and DES/SCF/PLE)

Traditional extraction of bioactive products from microalgae also employs different extraction strategies [52]. For example, chlorform/methanol or other mixed solvents can be used to extract pigments, followed by non-polar solvents such as hexane to extract lipids, and finally, proteins or polysaccharides can be extracted from the aqueous phase [40, 53]. Green extraction processes require reducing the use of toxic solvents while ensuring satisfactory extraction efficiency under new solvent conditions. To replace organic solvents such as methanol and ether,

Table 3 Extraction of lipids in marine microalgae

Microalgae	Compounds	Samples	Pretreatment	Extraction	Yield	Efficiency	Ref.
Phaeodactylum tricornutum	Lipids	Fresh	Freeze-drying + milling	Dimethyl Ether	5.8%	-	[102]
				Soxhlet	8.2%	-	
Nannochloropsis sp.	Lipids	Dried	Cholinium Arginate solution	Bligh and Dyer method	-	94.1%	[57]
Thraustochytrium sp.	Docosa-hexaenoic Acid-Rich Oil	Dried	1-decyl-3-methyl-imidazolium bis(2,4,4-trimethylpentyl)phosphinate	Hexane	69.4%	97.4%	[56]
N. oceanica	Polyunsatu-rated fatty acids	Spray-dried	Freeze-thaw cycles	10.5 MPa and 20 min, ethanol, 57 °C, 3 cycles→10.5 MPa and 20 min, limonene, 50 and 70 °C	100%	-	[66]
Nannochloropsis sp.	Lipids	Spray drying method	Lyophilisation, mortar and pestle	Ethyl lactate	31.1%	-	[103]
				Hexane	9.8%	-	
Nannochloropsis gaditana	Omega-3 acylglycerols and glycolipids	Wet microal-gal biomass	Enzyme pretreatment	Pressurized liquid extraction (hexane at 120 °C)	17.6%	-	[47]
			Without pretreatment		9.5%	-	
			Enzyme pretreatment	Folch method	16.9%	-	
			Without pretreatment		15%	-	
N. oceanica	Omega-3 fatty acids	Spray-dried biomass	-	Chloroform:Methanol (2:1, v/v)	28.7%	-	[70]
				Dichloromethane:Methanol (2:1, v/v)	29.0%	-	
				Dichloromethane:Ethanol (DE) (2:1, v/v)	11.9%	-	
				Ethanol	19.1%	-	
				Ethanol with ultrasonic bath	21.8%	-	
				Ethanol with ultrasonic probe	19.8%	-	
Chlorella sp.	Lipids	Dried biomass	-	Methanol and hexane (2:1 v/v) using an ultrasonic bath	31%	-	[35]
Schizochytrium DT3	Lipids	Freeze dried biomass	Bead milling	Chloroform: methanol (2:1)	49.4%	-	[45]
N. oceanica	Lipids	Wet microal-gal biomass	-	Ultrasound-assisted N, N, N′, N′-tetraethyl-1,3-propanediamine	-	98.2%	[60]
				Ultrasound-assisted Hexane	-	40.1%	
				Ultrasound-assisted Chloroform	-	51.9%	
				Ultrasound-assisted Chloroform-methanol	78.3%	-	
Nannochloropsis sp.	Lipids	Lyophilized biomass	Cellulase pretreatment	8:1 ethanol/biomass ratio (v/w) in 10–15% water content with 10% lipase weight	46.53–48.57%	-	[48]

Table 3 (continued)

Microalgae	Compounds	Samples	Pretreatment	Extraction	Yield	Efficiency	Ref.
Nannochloropsis oculata	Lipids	Freeze dried biomass	-	(hydroxymethyl) phosphonium chloride ionic liquid→ Methanol	12.8%	-	[104]
				Bligh & Dyer	17.3%	-	
				Soxhlet	9.1%	-	
Nannochloropsis sp.	Lipids	Lyophilized biomass	-	Hexane	49.4%	-	[105]
				Ethanol (96% v/v)	37.3%	-	
Thraustochytrium sp.	Lipids	Freeze dried biomass	1-ethyl-3-methylimidazolium ethylsulfate and the phosphonium tetrabutylphosphonium propanoate	Hexane	91%	-	[106]
Neochloris oleoabundans	Lipids	Wet biomass	Carbonate-heating treatment with N, N′ – dibutylurea	Ethanol-added aqueous two-phase solution	202.8 mg/g	97.9%	[107]
				Soxhlet	150 mg/g	-	
Nannochloropsis sp.	Lipids	Dried biomass	-	20 wt% ammonium sulfate, Tetramethylguanidine:menthol (3:1)	172 mg/g	-	[83]
N. gaditana	Lipids	Dried biomass	No pretreatment	Methanol-HCl solution → hexane	111.6 mg/g	-	[42]
			Ultrasonic method	Choline chloride: Ethylene glycol (1:2) →Methanol-HCl solution → hexane	130.9 mg/g		

-, none

numerous ionic liquids (ILs) or deep eutectic solvents (DESs) have been developed, or supercritical fluid (SCF) extraction techniques have been employed; to enhance extraction efficiency, technologies such as microwave, ultrasound, and pulsed electric fields have been integrated to promote the extraction efficiency of target active products.

Ionic liquids are composed entirely of cations and anions, featuring low vapor pressure, high solubility, and superior chemical and thermal stability compared to organic solvents. The physical properties of ionic liquids (polarity, hydrophobicity, and viscosity) can be controlled by combining different cations and anions, making them versatile for applications ranging from organic synthesis and catalytic reactions to electrochemistry, materials chemistry, separation technology, environmental science, and biomass energy [23]. In enhanced extraction, ionic liquids are used for enhanced extraction of pigments from higher plants, lipids and proteins from microalgae, and can be applied in enzyme-catalyzed reactions, lipid extraction [54], and the assisted- extraction of pigments [55]. Additionally, they possess reusability and are often referred to as green solvents. In terms of cell disruption, there are also reports on the use of ionic liquids. Literature has confirmed that the anions of ionic liquids can form stronger interactions with hydroxyl groups in microalgal cell walls than hydrogen bonds within the cell walls, thereby disrupting the cell walls [56]. Fifteen cholinium + amino acid-based ionic liquids (ILs) were evaluated to destroy the hydrogen-bond network in cell wall of *Nannochloropsis* sp. [57]. It was shown that cholinium arginate ([Ch][Arg]) was the best one for lipid extraction. The possible reason was [Ch][Arg] has 6 hydrogen bonding sites which was higher than those of the other fourteen ionic liquids. Shankar et al. [54] used hydrophilic ionic liquids combined with microwave-assisted extraction technology to process *N. oculata*, achieving a 34.9% increase in lipid yield compared to the classic chloroform-methanol system of Bligh-Dyer. More notably, this process exhibits superior mass transfer efficiency during wet algae processing, meaning that the energy-intensive algae drying pretreatment step can be omitted, which holds significant practical value for industrial-scale continuous production.

Compared to traditional organic solvent systems, functionalized solvent systems, ionic liquids and deep eutectic solvents, demonstrate multiple technical advantages in the field of microbial carotenoid enrichment, including enhanced selectivity, improved solventization capacity, and recyclability. ILs, based on organic cation-anion composite structures, have a low melting point of < 100 °C. Their low volatility, flame retardancy, and high thermal/chemical stability make them an important medium in green chemistry [58]. DES, as a new type of

Table 4 Extraction of pigments in marine microalgae

Microalgae	Compounds	Samples	Pretreatment	Extraction	Yield or content	Purity	Ref.
Nitzschia laevis.	*Fucoxanthinol and fucoxanthin*	Freeze-dried powders	Liquid nitrogen	Methanol→Solid-phase extraction →thin-layer chromatography	-	-	[76]
Phaeodactylum tricornutum	Fucoxanthin	Lyophilized microalgal powder	-	Methanol	7 mg/g	-	[108]
				70% ethanol	7 mg/g	-	
				Ethanol	3.5 mg/g	-	
Thalassiosira andamanica	Fucoxanthin, Chlorophyll, and Biosilica	Fresh	Frozen biomass	Benzyltriethyl ammonium chloride: Butane 1,4 diol (1:2) →petroleum ether and diethyl ether (PE: DEE, 2:1, v/v) in presence of water (7% w/w)	19.93 mg/g	-	[62]
				Metahnol	10.39 mg/g	-	
				Ethanol	13.05 mg/g	-	
Corynebacterium glutamicum	Astaxanthin	Fermentation Broth	-	90% ethanol	1.4 mg/g	-	[109]
				Acetone	0.88 mg/g	-	
				Ethyl acetate	1.1 mg/g	-	
Tetraselmis sp. CTP4	β-Carotenoid	Dried biomass	Glass beads	Tetrahydrofuran	2.3 mg/g	-	[37]
			Dispersion	Tetrahydrofuran	1.3 mg/g	-	
		Wet biomass	Glass beads	Tetrahydrofuran	3.0 mg/g	-	
				Methanol	2.5 mg/g	-	
			Dispersion	Tetrahydrofuran	2.5 mg/g	-	
	Lutein	Wet biomass	Dispersion	Acetate	0.2 mg/g	-	
				Tetrahydrofuran	0.4 mg/g	-	
			Glass beads	Acetate	0.47 mg/g	-	
				Tetrahydrofuran	0.62 mg/g	-	
Chlorella variabilis ATCC PTA 12,198	Lutein	Wet biomass	Glass beads	Tetrahydrofuran	3.2 mg/g	-	[79]
Muriellopsis sp. *(MCH35)*	Lutein	Spray-dried biomass	-	57 or 59 °C, 30 Mpa, 30% ethanol as a co-solvent	3.45 mg/g	-	[110]
		Freeze-dried biomass	-		4.20 mg/g	-	
P. tricornutum	Fucoxanthin	Freeze-dried biomass	-	20 MPa, 30 °C, and 30% ethanol	2.55%	11.07 mg/g	[111]
				20 MPa, 50 °C, and 30% ethanol	2.73%	8.56 mg/g	
				30 MPa, 30 °C, and 30% ethanol	3.00%	60.62 mg/g	
				30 MPa, 50 °C, and 30% ethanol	3.05	37.89 mg/g	
				40 MPa, 30 °C, and 30% ethanol	2.61	29.06 mg/g	

Table 4 (continued)

Microalgae	Compounds	Samples	Pretreatment	Extraction	Yield or content	Purity	Ref.
				40 MPa, 50 °C, and 30% ethanol	3.14	26.67 mg/g	
Chaetoceros calcitrans	Fucoxanthin	Lyophilized biomass	-	90% diallylammonium diallylcarbamate	17.51 mg/g	-	[55]
				Methanol	5 mg/g	-	
				Acetate	2 mg/g	-	

-, none

low-melting eutectic system, forms a low-melting eutectic system through intermolecular interactions between hydrogen bond acceptors (HBAs) and donors (HBDs), with a mixed melting point significantly lower than that of individual components [23]. Both solvent systems exhibit high design flexibility, allowing the construction of solvent systems with specific physicochemical parameters by adjusting the component ratios, thereby achieving efficient dissolution of various types of carotenoids. Notably, their raw materials are primarily derived from natural renewable resources, exhibiting low ecological toxicity, excellent biocompatibility, and biodegradability, providing technical support for sustainable production [58].

In recent years, the demand for efficient extraction of lipids and bioactive components from microalgal biomass has significantly increased, while stricter environmental regulations have driven the development of green solvents. Against this backdrop, novel systems with both biocompatibility and environmental friendliness have garnered significant attention, with ILs and DES emerging as research hotspots [59]. As an innovative alternative to traditional organic solvents, ionic liquids exhibit unique physicochemical properties. They are essentially molten salt systems composed of organic cations and inorganic/organic anions, with melting points generally below 100 °C (some remain liquid at room temperature). The core advantage of this solvent system lies in its structural designability: by adjusting the combination of cations and anions, key parameters such as polarity, thermal stability index, hydrophobicity, and viscosity coefficient can be precisely controlled, enabling customized extraction tailored to different target products. Notably, switchable solvents (SS), a functionalized branch of ILs, include two categories: polarity-switchable solvents (PSS) and hydrophilicity-switchable solvents (HSS). Their physical and chemical properties can be dynamically switched through external stimuli such as pH value and CO_2 concentration [60]. Although ILs show promising applications in green extraction, their environmental compatibility remains controversial: on one hand, some ILs have poor biodegradability; on the other hand, their synthesis may involve toxic precursors, complicating the assessment of their environmental impact across the entire lifecycle. It is worth noting that studies have validated the extraction efficiency of ILs in the field of microalgae biorefining.

Deep eutectic solvents (DES) and their natural deep eutectic solvents (nDES) have emerged as a new class of green solvents and have garnered significant attention in the field of sustainable chemistry in recent years. Their structural characteristics involve hydrogen bond acceptors (HBAs, such as quaternary ammonium salts like choline chloride) and hydrogen bond donors (HBDs, such

as biomass derivatives like urea and lactic acid) forming low-melting-point mixtures through hydrogen bond networks. Compared to ionic liquids and traditional organic solvents, this system exhibits multiple environmentally friendly attributes: biodegradability, ease of synthesis, and operational safety (non-volatile, non-flammable) make it a leading choice in the field of green chemistry. Notably, natural deep eutectic solvents (nDES) offer food-grade safety, allowing them to be directly applied in food systems after functional component extraction without requiring purification, significantly simplifying production processes. To address solvent separation challenges, researchers have developed responsive deep eutectic solvents (sDES) [61], which enable reversible regulation of solvent polarity through external stimuli such as pH and temperature, effectively resolving the technical bottleneck of separating target products from solvents.

Currently, the extraction of bioactive compounds from marine microalgae still relies on some flammable, explosive, volatile, and environmentally harmful solvents. Compared to green solvents, these traditional solvents offer higher extraction efficiency, lower production costs, and easier scalability. The environmental impact of green solvents also requires comprehensive consideration. It was proved that the photostability of Chlorophyll in DES (Benzyl-triethylammonium chloride: Caprylic Acid, 1:2) was higher than that in methanol shown in Table 4 [62].

The use of supercritical fluids for lipid extraction is an emerging green technology that has recently gained significant attention and recognition for its application in microalgae processing. It has been introduced as an alternative to organic solvents. Carbon dioxide is one of the most widely used solvents of SCFs for lipid extraction (Table 5). It is a high equipment-cost, non-toxic solvent that avoids degradation of extracted components due to high temperatures, as its critical temperature is close to room temperature [63–65]. However, its main drawback is its low polarity, which results in lower extraction efficiency for certain components. The lipid yields for *P. tricornutum,* and *N. oculata* were more than 10% but that was less than 5% for *Porphyridium cruentum* [63] (Table 5). Polar co-solvents can be used to compensate for this deficiency. Other supercritical fluids that can be used in the lipid extraction process include methanol and ethanol [66, 67].

Pressurized liquid extraction (PLE) is a multifaceted modern extraction technology with alternative names such as PFE and Accelerated Solvent Extraction (ASE). Compared to traditional extraction methods, PLE demonstrates multi-dimensional technical advantages: enhanced selective control capability, over 50% reduction in processing time, and significantly reduced solvent consumption. Extraction pressure: 103.4 bar, 100% dimethyl sulfoxide (DMSO), yielding 0.82 mg/L lutein, 2.84 β-carotene, 33.12 mg/L fucoxanthin, 2.28 mg/L diatoxanthin [68], and fatty acids, etc. [29].

Process intensification & hybrid approaches

Microwave technology is based on the energy conversion mechanism of electromagnetic waves in the 0.3–300 GHz frequency band. Studies have shown that this technology combines rapid response advantages, and has been successfully applied to the pretreatment and efficient extraction of various bioactive substances such as microalgal polysaccharides, pigments, and lipids, and is recognized as a green extraction technology with industrialization potential. To extract lutein from the marine microalgae *Chlorella sorokiniana* (NIOT-2), microwave treatment for 1.47 minutes, a solid-liquid ratio of 36.8:1 (mg/mL), and the addition of 8.16 M KOH yield the optimal pretreatment effect [69].

Ultrasound-assisted extraction is another technique used to disrupt cell walls and enhance extraction efficiency. The lipid yield using ultrasonic bath was higher than those of using ultrasonic probe and without ultrasound when ethanol was performed as the solvent shown [70]. Ethanol is less toxic than methanol, dichloromethane and chloroform, but ethanol has lower extraction yield (Table 3). It is indicated that green extraction is not easy and alternative solvents should be developed. The increase in extraction yield is attributed to enhanced mass transfer caused by microfluidics, cavitation, and bubble rupture, leading to cell disruption. Ultrasonic power of 200 W or 300 W, and ultrasonication time of 20 or 25 minutes can both break down the cell walls of *Tetraselmis subcordiformis* and effectively extract starch [39]. The ultrasonication process must also control temperature to prevent overheating. Currently, it is also limited by scale. For marine *Chlorella* sp. NITT 02, wet algae are added directly to an ammonium sulfate aqueous solution without drying, followed by ultrasonic treatment. Tert-butanol is then added to form a three-phase system: the top layer is a tert-butanol layer containing oils; the middle layer is the protein layer; and the bottom layer is the sugar-containing salt solution layer [71]. Pulse electric field-assisted extraction (PEFAE) is a novel physical field separation technology that induces cell membrane electroporation through high-intensity pulse electric fields to achieve non-thermal cell wall disruption. However, the pretreatment effect of the pulse electric field is inferior to that of ultrasonication shown in Table 2 [41].

Microwaves and ultrasonication can not only assist in cell wall disruption but can also be coupled with the extraction process to enhance the extraction of bioactive compounds from microalgae. The primary function of ultrasound is also to enhance cell disruption for the lipid extraction from fresh *N. oceanica* [60]. It was not an

Table 5 Extraction of active components in marine microalgae by supercritical fluids

Microalgae	Compounds	Samples	Pretreatment	Extraction	Yield	Ref.
P. tricornutum	Polyunsaturated fatty acids	Dried biomass	-	300 bar, 40 °C, CO_2; n-butane at 15 bar, 40 °C	14.7 (CO_2) and 12.8 (n-butane)	[63]
Nannochloropsis oculata					15.6 (CO_2) and 17.2 (n-butane)	
Porphyridium cruentum					4.5 (CO_2) and 2.4 (n-butane)	
N. oceanica	Carotenoids	Spray-dried	-	300 MPa, ethanol, 50 °C →100 MPa, limonene, 50 °C	115.1 mg/g	[66]
Nannochloropsis sp.	Lipids	Lyophilized biomass	-	35 MPa, 50 ∘C, 8 g CO2/min + 10 wt% ethanol, 8 h→7 MPa, 50 ∘C, 8 g CO_2/min + 70 wt% ethanol, 1 h	25.9% → 13.2%	[67]
				Butyl acetate	87.5%	
Nannochloropsis gaditana	Lipids	Lyophilized biomass	Mechanical pre-treatment	65 °C and 250 bars with a CO_2	27.4% EPA	[64]
Nannochloropsis salina and Nannochloropsis maritime	Lipids and pigments	Dried biomass	-	CO_2, 333 K and 300–400 bar	14%	[65]
Tetraselmis chuii	Lipids	Freeze-dried biomass	Freeze/thaw cycles + ultrasound waves	15 MPa, 40 °C, CO_2	2.5%	[112]
			Without ultrasound waves		0.6%	
Schizochytrium sp.	Lipids	Lyophilised biomass	-	55 °C and 70 MPa, CO_2	347.17×10^{-6} (Maximum extraction rate,/s)	[113]
				Soxhlet extraction	17.02×10^{-6} (Maximum extraction rate,/s)	

-, none

independent process. Although the ultrasound Wahidin et al. [72] integrated microwave technology, ionic liquids, and organic solvents to achieve direct transesterification of lipid components in *Nannochloropsis* sp. Among these, the system using 1-ethyl-3-methylimidazolium methyl sulfate with methanol demonstrated significantly higher lipid extraction efficiency compared to systems using hexane, chloroform, 1-butyl-3-methylimidazolium chloride, and 1-butyl-3-methylimidazolium trifluoromethane sulfonate with methanol. If freezing + thawing and ultrasonication were integrated, the extraction yield of C-Phycocyanin was 109.57 mg/g which was higher than that of freezing + thawing or ultrasonication [34]. At the same time, the purity of freezing + thawing and ultrasonication was the highest compared with freezing + thawing, ultrasonication, maceration or homogenization shown in Table 2.

Alternatively, an electric field can be used to enhance the water extraction of proteins, polysaccharides, and pigments, followed by extraction of fats using 95% ethanol or chloroform/methanol (2/1, v/v) [73].

These extraction methods are compared in Table 6. Extraction methods based on ionic liquids and deep eutectic solvents require the development of inexpensive ionic liquids to reduce production costs and solid-liquid ratios, thereby enhancing the feasibility of large-scale production.

Downstream purification, scale-up and LCA

It should be noted that most active products require further purification via ultrafiltration [34, 74, 75] or thin-layer chromatography [76] after extraction. In general, distillation is traditional method for the recovery of organic solvents. Membrane technology was selected to purify proteins from solutions.

To improve the technical and economic viability of microalgae active product development, it is necessary to develop multiple products. One goal of microalgae biomass product separation is to achieve the full utilization of microalgae biomass resources as much as possible. Currently, there is limited research on multi-product separation [77]. Based on separation principles, a single separation process may yield a purified component from a multi-component mixture, with the final unit operation producing two purified components. Given the numerous components in microalgae, achieving complete separation would require a large number of unit operations. Therefore, it is essential to rationally design extraction processes based on separation principles and select environmentally friendly solvents. On the other hand, the economic value of these microalgal components varies greatly. Therefore, from an economic perspective, it is not necessary to achieve complete separation of microalgal products. It is sufficient to obtain several products with the highest economic value, while the separation residues can be processed through thermal conversion or fermentation [78], thereby achieving resource recycles. The cultivation of marine microalgae can be integrated with wastewater treatment. The further utilization of the residual biomass after extraction is also important. These processes are also critical for blue biotechnology.

Most current research is based on small-scale separation operations at the laboratory level, typically involving volumes of a few dozen milliliters. However, it is now necessary to study separation processes at the liter scale. Only by obtaining separation results from pilot-scale or actual production processes can reliable assessments of process economics and environmental safety be conducted. Currently, compounds such as ionic liquids with high separation efficiency are still relatively expensive. To reduce costs, extraction solvents need to be recycled. The recovery and reprocessing of used solvents may also have environmental impacts. For example, some compounds are highly stable and may be considered non-biodegradable persistent pollutants. The environmental impact of compounds is complex; some compounds are non-toxic or low-toxic to most organisms, but long-term environmental assessment data are lacking. Therefore, the rational design of the green solvents and eco-toxicity analysis of these solvents are essential target for the development of green extraction of bioactive compounds in marine microalgae. The scale-up production of ILs and reduction

Table 6 Comparison of extraction methods for bio-products in marine microalgae

Method		Equipment cost	Reagent cost	Energy cost	Eco-toxicity	Scalability	Product stability	Yield
Solvent-based	Traditional organic solvent	Cheap	Cheap	Low	High	Easy	Moderate	Moderate to high
	Ionic liquids and deep eutectic solvents	Cheap	Expensive	Low	Low	Difficult	High	High
	Supercritical fluid extraction	Expensive	Cheap	High	Low	Moderate	High	High
	Pressurized liquid extraction	Expensive	Cheap	High	Low	Moderate	High	High
Non-solvent-based	Enzyme	Cheap	Expensive	Low	Low	Easy	High	High
	Ultrasound	Expensive	No	High	Low	Difficult	High	High
	Microwave	Expensive	No	High	Low	Difficult	Low	High

of production cost of ILs are also important. As previously mentioned, the biorefining process of microalgae involves multiple steps, with downstream processes being particularly complex. A pilot test is based on the laboratory scale results and the scale-up need all the informations of equipment, operation parameters and the analysis methods of final products and by-products. To achieve greener development of microalgal bioactive compounds, beyond solvent selection, harvesting, separation, and purification processes all require further optimization to reduce energy consumption and environmental pollution. Extracting oils from marine *Chlorella* algae, Wet Algae has clearly omitted the freeze-drying process, resulting in lower energy consumption [35]. Beyond freeze-drying (1080 MJ), ultrasonic disruption (108–324 MJ) and solvent recovery (210–588 MJ) are also energy-intensive processes, significantly more so than harvesting (0.02 MJ) [35]. Sadukha et al. [79] designed a multi-product separation technology roadmap for marine *C. variabilis* and conducted analyses of raw material costs and energy consumption. Among these processes, sugar extraction required high temperatures and sulfuric acid, resulting in the highest energy consumption (160.8 MJ/kg). Cell disruption (36.0 MJ/kg) and protein extraction (19.3 MJ/kg) followed as the next most energy-intensive steps, while harvesting (5.2 MJ/kg), lutein and phytol extraction (7.2 MJ/kg), and lipid extraction (5.9 MJ/kg) all exhibited relatively low energy demands. LCA is a methodological framework that is powerful to investigate the potential environmental performance of biorefinery and bioproduct development of marine microalgae. The LCA includes four steps: defining the aim and scope, life cycle inventory analysis, impact assessment and interpretation phase. The standard procedure and the minimum data set for LCA were shown in references [80–82].

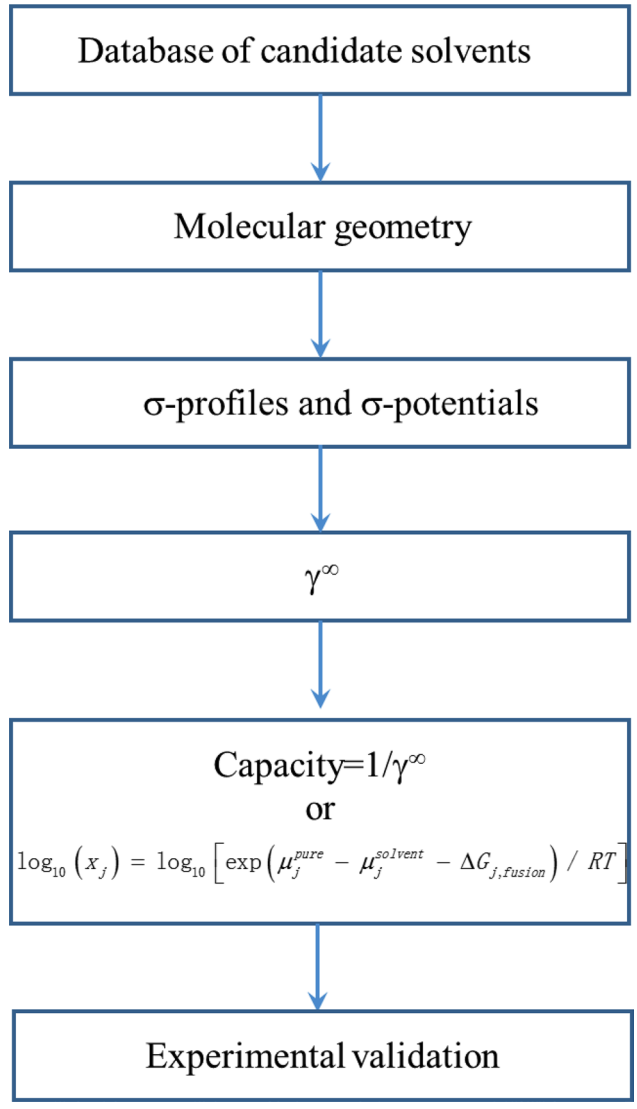

Fig. 2 Procedure of solvent selection by COSMO-RS

Computational tools & AI

The extraction of active products from marine microalgae follows thermodynamic principles. Based on the different phase compositions of the separation process, it can be categorized into solid-liquid extraction [76], liquid-phase extraction (including two-phase liquid extraction [39, 74]), and three-phase extraction [83], among others. Since data on solid-liquid equilibrium, liquid-liquid equilibrium, and other parameters are primarily concentrated on ternary and quaternary systems, there are still challenges in selecting suitable solvents. Currently, theoretical calculations such as COSMO-RS [84] can provide some theoretical guidance. However, actual separation systems are typically complex multi-component systems that also include components such as salts, ionic liquids, and polymers, making it challenging to predict the effects of phase separation. The procedure of solvent selection by COSMO-RS is shown in Fig. 2.

Microalgae component separation is based on principles such as liquid-liquid phase equilibrium or solid-liquid phase equilibrium. Current experimental studies primarily focus on simple systems, but microalgae contain multiple components that may influence separation behaviour. However, data for ternary or quaternary systems are incomplete, and theoretical guidance cannot be provided for most separation processes. Although theoretical calculation tools such as COSMO-RS exist, obtaining stable compound structures through calculation is time-consuming and labor-intensive for complex systems, let alone phase equilibrium separation calculations involving multiple components. Additionally, the physical and chemical properties of compounds, such as density and viscosity, also affect the separation process

and require thermodynamic measurements or theoretical estimates. Until now, COSMO-RS has been used for the extractions of lutein [85], phycobiliprotein [86], EPA [87], fucoxanthin [88] and lipids [89]. Although simulation is easier than experiment, there are lots of possible solvents. About 24 [88] and [85, 90] candidate DESs were proposed, respectively. The final solvents were selected by COSMO-RS for them. About 8,000 molecules were collected and databases were constructed for screening the solvents [91, 92]. Additional constraints of boiling point, melting point, overall greenness score, and solubility, the number of the candidate solvents decreased to 712 [91].

As mentioned earlier, there is currently insufficient thermodynamic data on liquid-liquid phase equilibrium and solid-liquid phase equilibrium for the extraction of active products from microalgae, especially for multi-component systems, which poses challenges in solvent screening and other areas. While theoretical calculations can provide some guidance, the computational demands of extensive solvent screening are time-consuming and may not align with actual systems. There is a wealth of existing laboratory separation research that can be analyzed using machine learning [93, 94] or deep learning algorithms to explore solvent screening and separation condition design for specific systems. Based on the results of these artificial intelligence analyses, experimental validation can be conducted to establish a cycle of experimental science, theoretical calculations, and artificial intelligence analysis, thereby revealing the intrinsic patterns and potential technical routes for the extraction of multiple products from microalgae. The green extraction process for the active components of microalgae can also be referenced for the extraction of active products from marine macroalgae, marine microorganisms, or animals, thereby promoting the development of marine biotechnology—blue biotechnology.

To achieve large-scale green extraction of bioactive compounds from marine microalgae, numerous challenges remain to be addressed. It is necessary to develop standardized pilot-scale extraction test suites; quantify solvent life-cycle impacts; create open databases of extraction yields for ML training.

Conclusions

To obtain active substances from microalgae, the entire process must be optimized, including sample drying or selection of wet algae, sample pretreatment, extraction, and purification. Traditional organic solvent extraction is highly efficient, but newly developed green processes still face numerous challenges that need to be addressed, such as improving extraction efficiency and reducing production costs. Process intensification technologies like microwave or ultrasound have high energy consumption or are limited in processing scale. Theoretical computational studies also require further refinement. This involves starting with the analysis of microalgal cell wall structures, selecting extraction solvents from a phase equilibrium perspective, and integrating new technologies such as artificial intelligence to drive the optimization of green extraction processes. Green, high efficient, and sustainable extraction of bioactive compounds from marine microalgae will drive the large-scale application of blue biotechnology.

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No datasets were generated or analysed during the current study.

Materials availability
All the figures are prepared by the authors, unnecessary for the permission of reproduced images.

Declarations

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References

1. Tschitschko B, Esti M, Philippi M, Kidane AT, Littmann S, Kitzinger K, et al. Rhizobia-diatom symbiosis fixes missing nitrogen in the ocean. Nature. 2024;630(8018):899–904.
2. Castro-Ferreira C, Gomes-Dias JS, Ferreira-Santos P, Pereira RN, Vicente AA, Rocha CMR. *Phaeodactylum tricornutum* extracts as structuring agents for food applications: physicochemical and functional properties. Food Hydrocolloids. 2022;124:107276.
3. Kaur S, Sharma A, Bala S, Satheesh N, Nile AS, Nile SH. Microalgae in the food-health nexus: exploring species diversity, high-value bioproducts, health benefits, and sustainable market potential. Bioresour Technol. 2025;427:132424.
4. Puri M, Gupta A, McKinnon RA, Abraham RE. Marine bioactives: from energy to nutrition. Trends Biotechnol. 2022;40(3):271–80.
5. Narayanan M. Marine algae biomass. A viable and renewable resource for biofuel production: a review. Algal Res. 2024;82:103687.
6. Jawaji A, Zilberg D, Khozin-Goldberg I. Using byproduct of fucoxanthin production by *Phaeodactylum tricornutum* for the development of a treatment against monogenean infection in fish. J Appl Phycol. 2023;35(6):2719–30.

7. Tang DYY, Ooi C, Show PL. Blue bioeconomy and biotechnology: towards a sustainably growing microalgae industry. Blue Biotechnol. 2025;2(1):13.
8. Diankristanti PA, Ng IS. Marine microalgae for bioremediation and waste-to-worth valorization: recent progress and future prospects. Blue Biotechnol. 2024;1(1):10.
9. Patel AK, Vadrale AP, Tseng Y-S, Chen C-W, Dong C-D, Singhania RR. Bioprospecting of marine microalgae from Kaohsiung Seacoast for lutein and lipid production. Bioresour Technol. 2022;351:126928.
10. Benbelkhir FZ, Medjekal S. Microalgal carotenoids: a promising alternative to synthetic dyes. Algal Res. 2022;66:102823.
11. Zhuang G, Ye Y, Zhao J, Zhou C, Zhu J, Li Y, et al. Valorization of *Phaeodactylum tricornutum* for integrated preparation of diadinoxanthin and fucoxanthin. Bioresour Technol. 2023;385:129412.
12. Ouedrhiri W, Bennis I, El Arroussi H. Recent advances in microalgae-based vitamin D metabolome: biosynthesis, and production. Bioresour Technol. 2024;407:131078.
13. Seth K, Kumar A, Rastogi RP, Meena M, Vinayak V, Harish. Bioprospecting of fucoxanthin from diatoms-challenges and perspectives. Algal Res. 2021;60:102475.
14. Yoshida E, Kato Y, Kanamoto A, Kondo A, Hasunuma T. Mixotrophic culture enhances fucoxanthin production in the haptophyte *Pavlova gyrans*. Appl Microbiol Biotechnol. 2024;108(1):352.
15. Kabilan C, Roy RK, Chadha A. Docosahexaenoic acid production by a novel high yielding strain of *Thraustochytrium* sp of Indian origin: isolation and bioprocess optimization studies. Algal Res. 2018;32:93–100.
16. Mitra M, Patidar SK, George B, Shaha F, Mishra S. A euryhaline *Nannochloropsis gaditana* with potential for nutraceutical (EPA) and biodiesel production. Algal Res. 2015;8:161–67.
17. Popovich CA, Faraoni MB, Sequeira A, Daglio Y, Martin LA, Martinez AM, et al. Potential of the marine diatom *Halamphora coffeaeformis* to simultaneously produce omega-3 fatty acids, chrysolaminarin and fucoxanthin in a raceway pond. Algal Res. 2020;51:102030.
18. Magoni C, Bertacchi S, Giustra CM, Guzzetti L, Cozza R, Ferrari M, et al. Could microalgae be a strategic choice for responding to the demand for omega-3 fatty acids? A European perspective. Trends Food Sci Techol. 2022;121:142–55.
19. Chandra R, Iqbal HMN, Vishal G, Lee HS, Nagra S. Algal biorefinery: a sustainable approach to valorize algal-based biomass towards multiple product recovery. Bioresour Technol. 2019;278:346–59.
20. Derwenskus F, Weickert S, Lewandowski I, Schmid-Staiger U, Hirth T. Economic evaluation of up- and downstream scenarios for the co-production of fucoxanthin and eicosapentaenoic acid with *P. tricornutum* using flat-panel airlift photobioreactors with artificial light. Algal Res. 2020;51:102078.
21. Jothibasu K, Muniraj I, Jayakumar T, Ray B, Dhar DW, Karthikeyan S, et al. Impact of microalgal cell wall biology on downstream processing and nutrient removal for fuels and value-added products. Biochem Eng J. 2022;187:108642.
22. de Carvalho Jc, Magalhaes Ai DMPG Jr, Medeiros ABP, Sydney EB, Rodrigues C, Aulestia DTM, et al. Microalgal biomass pretreatment for integrated processing into biofuels, food, and feed. Bioresour Technol. 2020;300:122719.
23. Raj T, Morya R, Chandrasekhar K, Kumar D, Soam S, Kumar R, et al. Microalgae biomass deconstruction using green solvents: challenges and future opportunities. Bioresour Technol. 2023;369:128429.
24. Couto D, Conde TA, Melo T, Neves B, Costa M, Cunha P, et al. Effects of outdoor and indoor cultivation on the polar lipid composition and antioxidant activity of *Nannochloropsis oceanica* and *Nannochloropsis limnetica*. A lipidomics perspective. Algal Res. 2022;64:102718.
25. Renteria-Mexia A, Ulloa-Mercado G, Gortares-Moroyoqui P, Gonzalez-Mercado A, Sanchez M, Sineiro J, et al. Antioxidant potential and antiangiogenic activity of *Tetraselmis suecica* grown in a semicontinuous culture. J Chem Technol Biotechnol. 2022;97(9):2528–36.
26. Chua ET, Dal'molin C, Thomas-Hall S, Netzel ME, Netzel G, Schenk PM. Cold and dark treatments induce omega-3 fatty acid and carotenoid production in *Nannochloropsis oceanica*. Algal Res. 2020;51:102059.
27. Parra-Riofrio G, Garcia-Marquez J, Casas-Arrojo V, Uribe-Tapia E, Abdala-Diaz RT. Antioxidant and cytotoxic effects on tumor cells of exopolysaccharides from *Tetraselmis suecica* (Kylin) butcher grown under autotrophic and heterotrophic conditions. Mar Drugs. 2020;18(11):534.
28. Serra AT, Silva SD, de Gouveia LP, Alexandre AMRC, Pereira CV, Pereira AB, et al. A single dose of marine *Chlorella vulgaris* increases plasma concentrations of lutein, beta-carotene and zeaxanthin in healthy male volunteers. Antioxidants. 2021;10(8):1164.
29. Zhou J, Wang M, Bauerl C, Cortes-Macias E, Calvo-Lerma J, Carmen Collado M, et al. The impact of liquid-pressurized extracts of *Spirulina, Chlorella* and *Phaedactylum tricornutum* on in vitro antioxidant, antiinflammatory and bacterial growth effects and gut microbiota modulation. Food Chem. 2023;401:134083.
30. Banskota AH, Sperker S, Stefanova R, McGinn PJ, O'Leary SJB. Antioxidant properties and lipid composition of selected microalgae. J Appl Phycol. 2019;31(1):309–18.
31. Foo SC, Yusoff FM, Ismail M, Basri M, Chan KW, Khong NMH, et al. Production of fucoxanthin-rich fraction (FxRF) from a diatom, *Chaetoceros calcitrans* (Paulsen) Takano 1968. Algal Res. 2015;12:26–32.
32. Azizan A, Bustamam MSA, Maulidiani M, Shaari K, Ismail IS, Nagao N, et al. Metabolite profiling of the microalgal diatom *Chaetoceros Calcitrans* and correlation with antioxidant and nitric oxide inhibitory activities via H-1 NMR-based metabolomics. Mar Drugs. 2018;16(5):154.
33. Cui Y, Thomas-Hall SR, Chua ET, Schenk PM. Development of a *Phaeodactylum tricornutum* biorefinery to sustainably produce omega-3 fatty acids and protein. J Clean Prod. 2021;300:126839.
34. Tavanandi HA, Mittal R, Chandrasekhar J, Raghavarao KSMS. Simple and efficient method for extraction of C-Phycocyanin from dry biomass of *Arthospira platensis*. Algal Res. 2018;31:239–51.
35. Amin M, Chetpattananondh P. Enhanced lipid recovery from marine *Chlorella* sp. by ultrasonication with an integrated process approach for wet and dry biomass. Bioenerg Res. 2019;12(3):665–79.
36. Bezerra PQM, Kuntzler SG, Morais MG, Nunes IL, Costa JAV, Laroche C. Innovative strategy in the production of polysaccharides from *Spirulina* and *chlorella* grown in seawater and brackish groundwater. Bioenerg Res. 2024;17(3):1876–84.
37. Schuler LM, Gangadhar KN, Duarte P, Placines C, Molina-Marquez AM, Leon-Banares R, et al. Improvement of carotenoid extraction from a recently isolated, robust microalga, *Tetraselmis* sp. CTP4 (Chlorophyta). Bioproc Biosyst Eng. 2020;43(5):785–96.
38. Magpusao J, Giteru S, Oey I, Kebede B. Effect of high pressure homogenization on microstructural and rheological properties of *A. platensis, isochrysis, Nannochloropsis* and *Tetraselmis* species. Algal Res. 2021;56:102327.
39. Liao L, Shen Y, Xie C, Zhang Y, Yao C. Ultrasonication followed by aqueous two-phase system for extraction, on-site modification and isolation of microalgal starch with reduced digestibility. Ultrason Sonochem. 2024;106:106891.
40. Motlagh SR, Khezri R, Etesami M, Chee CY, Kheawhom S, Nootong K, et al. Microwave-assisted extraction of lipid and eicosapentaenoic acid from the microalga *Nanochloropsis* sp. using imidazolium-based ionic liquids as an additive in water. J Appl Phycol. 2024;36(4):1709–24.
41. Sommer M-C, Balazinski M, Rataj R, Wenske S, Kolb JF, Zocher K. Assessment of phycocyanin extraction from *Cyanidium caldarium* by spark discharges, compared to freeze-thaw cycles, sonication, and pulsed electric fields. Microorganisms. 2021;9(7):1452.
42. Martinez PM, Ortiz-Martinez VM, Segado SS, Salar-Garcia MJ, APdl R, Fernandez FJH, et al. Deep eutectic solvents for the extraction of fatty acids from microalgae biomass: recovery of omega-3 eicosapentaenoic acid. Sep Purif Technol. 2022;300:121842.
43. Ma NL, Teh KY, Lam SS, Kaben AM, Cha TS. Optimization of cell disruption methods for efficient recovery of bioactive metabolites via NMR of three freshwater microalgae (Chlorophyta). Bioresour Technol. 2015;190:536–42.
44. Sanjeewa KKA, Khinm H, Kim Y-S, Jeon Y-J, Kim S-K. Enzyme-assisted extraction of bioactive compounds from seaweeds and microalgae. trac-Trend Anal Chem. 2023;167:117266.
45. Byreddy AR, Barrow CJ, Puri M. Bead milling for lipid recovery from thraustochytrid cells and selective hydrolysis of *Schizochytrium* DT3 oil using lipase. Bioresour Technol. 2016;200:464–69.
46. Delran P, Barthe L, Peydecastaing J, Pontalier PY, Guiheneuf F, Frances C. Integrating wet stirred-bead milling for *Tetraselmis suecica* biorefinery: operating parameters influence and specific energy efficiency. Bioresour Technol. 2024;394:130181.
47. Castejon N, Javier Senorans F. Simultaneous extraction and fractionation of omega-3 acylglycerols and glycolipids from wet microalgal biomass of *Nannochloropsis gaditana* using pressurized liquids. Algal Res. 2019;37:74–82.
48. He Y, Wang X, Wei H, Zhang J, Chen B, Chen F. Direct enzymatic ethanolysis of potential *Nannochloropsis* biomass for co-production of sustainable biodiesel and nutraceutical eicosapentaenoic acid. Biotechnol Biofuels. 2019;12:78.
49. Maffei G, Bracciale MP, Broggi A, Zuorro A, Santarelli ML, Lavecchia R. Effect of an enzymatic treatment with cellulase and mannanase on the

structural properties of *Nannochloropsis* microalgae. Bioresour Technol. 2018;249:592–98.
50. Vishwakarma R, Malik A. Partial enzymatic cell wall disruption of *Oocystis* sp. for simultaneous cultivation and extraction. Sep Purif Technol. 2022;293:121107.
51. Al-Zuhair S, Ashraf S, Hisaindee S, Al Darmaki N, Battah S, Svistunenko D, et al. Enzymatic pre-treatment of microalgae cells for enhanced extraction of proteins. Eng Life Sci. 2017;17(2):175–85.
52. Singh D, Puri M, Wilkens S, Mathur AS, Tuli DK, Barrow CJ. Characterization of a new zeaxanthin producing strain of *Chlorella saccharophila* isolated from new Zealand marine waters. Bioresour Technol. 2013;143:308–14.
53. Motlagh SR, Harun R, Biak DRA, Hussain SA, Omar R, Khezri R, et al. Ionic liquid-based microwave-assisted extraction of lipid and eicosapentaenoic acid from *Nannochloropsis oceanica* biomass: experimental optimization approach. J Appl Phycol. 2021;33(4):2015–29.
54. Shankar M, Chhotaray PK, Gardas RL, Tamilarasan K, Rajesh M. Application of carboxylate protic ionic liquids in simultaneous microalgal pretreatment and lipid recovery from marine *Nannochloropsis* sp. and *Chlorella* sp. Biomass Bioenerg. 2019;123:14–24.
55. Khoo KS, Ooi CW, Chew KW, Foo SC, Show PL. Bioprocessing of *Chaetoceros calcitrans* for the recovery of fucoxanthin using CO_2-based alkyl carbamate ionic liquids. Bioresour Technol. 2021;322:124520.
56. Zhang Y, Ward V, Meramo S, Dennis D, Plechkova V, Armenta N, et al. Extraction of a docosahexaenoic acid-rich oil from *Thraustochytrium* sp. using a hydrophobic ionic liquid. ACS Sustain Chem Eng. 2023;11:5726–36.
57. Chua ET, Brunner M, Atkin R, Eltanahy E, Thomas-Hall SR, Schenk PM. The ionic liquid cholinium arginate is an efficient solvent for extracting high-value *Nannochloropsis* sp. lipids. ACS Sustain Chem. Eng. 2019;7(2):2538–44.
58. Sanches MV, Freitas R, Oliva M, Mero A, De Marchi L, Cuccaro A, et al. Are natural deep eutectic solvents always a sustainable option? A bioassay-based study. Environ Sci Pollut Res. 2023;30(7):17268–79.
59. Lo C, Wijffels RH, Eppink MHM. Lipid extraction from fresh *Nannochloropsis oceanica* using semi-hydrophobic eutectic solvents. Algal Res. 2023;72:103117.
60. Guo H, Cheng J, Mao Y, Qian L, Yang W, Park J-Y. Synergistic effect of ultrasound and switchable hydrophilicity solvent promotes microalgal cell disruption and lipid extraction for biodiesel production. Bioresour Technol. 2022;343:126087.
61. Zhang J, Li S, Yao L, Yi Y, Shen L, Li Z, et al. Responsive switchable deep eutectic solvents: a review. Chin Chem. Lett. 2023;34(5):107750.
62. Singh K, Paidi MK, Kulshrestha A, Bharmoria P, Mandal SK, Kumar A. Deep eutectic solvents based biorefining of value-added chemicals from the diatom *Thalassiosira andamanica* at room temperature. Sep. Purif Technol. 2022;298:121636.
63. Feller R, Matos AP, Mazzutti S, Moecke EHS, Tres MV, Derner RB, et al. Polyunsaturated ω-3 and ω-6 fatty acids, total carotenoids and antioxidant activity of three marine microalgae extracts obtained by supercritical CO_2 and subcritical n-butane. J Supercrit Fluid. 2018;133:437–43.
64. Molino A, Martino M, Larocca V, Di Sanzo G, Spagnoletta A, Marino T, et al. Eicosapentaenoic acid extraction from *Nannochloropsis gaditana* using carbon dioxide at supercritical conditions. Mar Drugs. 2019;17(2):132.
65. Mouahid A, Seengeon K, Martino M, Crampon C, Kramer A, Badens E. Selective extraction of neutral lipids and pigments from *Nannochloropsis salina* and *Nannochloropsis maritima* using supercritical CO_2 extraction: effects of process parameters and pre-treatment. J Supercrit Fluid. 2020;165:104934.
66. Gallego R, Bueno M, Chourio AM, Ibanez E, Saldana MDA, Herrero M. Use of high and ultra-high pressure based-processes for the effective recovery of bioactive compounds from *Nannochloropsis oceanica* microalgae. J Supercrit Fluid. 2021;167:105039.
67. Jimenez Callejon MJ, Robles Medina A, Macias Sanchez MD, Gonzalez Moreno PA, Navarro Lopez E, Esteban Cerdan L, et al. Supercritical fluid extraction and pressurized liquid extraction processes applied to eicosapentaenoic acid-rich polar lipid recovery from the microalga *Nannochloropsis* sp. Algal Res. 2022;61:102586.
68. Wang M, Moron-Ortiz A, Zhou J, Benitez-Gonzalez A, Mapelli-Brahm P, Melendez-Martinez AJ, et al. Effects of pressurized liquid extraction with dimethyl sulfoxide on the recovery of carotenoids and other dietary valuable compounds from the microalgae *Spirulina*, *Chlorella* and *Phaeodactylum tricornutum*. Food Chem. 2023;405:134885.
69. Mary Leema JT, Persia Jothy T, Dharani G. Rapid green microwave assisted extraction of lutein from *Chlorella sorokiniana* (NIOT-2) - process optimization. Food Chem. 2021;372:131151.
70. Figueiredo ARP, da Costa E, Silva J, Rosario Domingues M, Domingues P. The effects of different extraction methods of lipids from *Nannochloropsis oceanica* on the contents of omega-3 fatty acids. Algal Res. 2019;41:101556.
71. Anto S, Premalatha M, Mathimani T. Extraction and optimization of biomolecules recovery from *Chlorella* sp. NITT 02: unleashing potential through ultrasound and triphasic system of extraction. J Appl Phycol. 2024;36(5):2607–23.
72. Wahidin S, Idris A, Shaleh SRM. Ionic liquid as a promising biobased green solvent in combination with microwave irradiation for direct biodiesel production. Bioresour Technol. 2016;206:150–54.
73. Zhang R, Marchal L, Lebovka N, Vorobiev E, Grimi N. Two-step procedure for selective recovery of bio-molecules from microalga *Nannochloropsis oculata* assisted by high voltage electrical discharges. Bioresour Technol. 2020;302:122893.
74. Santos JHPM, Trigo JP, Maricato E, Nunes C, Coimbra MA, Ventura SPM. Fractionation of *Isochrysis galbana* proteins, arabinans, and glucans using ionic-liquid-based aqueous biphasic systems. ACS Sustain Chem. Eng. 2018;6(11):14042–53.
75. Nistico DM, Piro A, Oliva D, Osso V, Mazzuca S, Faga FA, et al. A combination of aqueous extraction and ultrafiltration for the purification of phycocyanin from *Arthrospira maxima*. Microorganisms. 2022;10(2):308.
76. Sun P, Wong C-C, Li Y, He Y, Mao X, Wu T, et al. A novel strategy for isolation and purification of fucoxanthinol and fucoxanthin from the diatom *Nitzschia laevis*. Food Chem. 2019;277:566–72.
77. Lam Gp T, Vermue MH, Eppink MHM, Wijffels RH, van den Berg C. Multi-product microalgae biorefineries: from concept towards reality. Trends Biotechnol. 2018;36(2):216–27.
78. Sinzinger K, Bieringer S, Schieder D, Riepl H, Sieber V. Biorefinery concept for *Cylindrospermum alatosporum* CCALA 988 extracting multiple high-value compounds and residue utilization by *P. pastoris* fermentation producing phytase. Algal Res. 2023;76:103302.
79. Sadukha S, Mehta B, Chatterjee S, Ghosh A, Dineshkumar R. Sequential downstream process for concurrent extraction of lutein, phytol, and biochemicals from marine microalgal biomass as a sustainable biorefinery. ACS Sustain Chem. Eng. 2023;11(2):547–58.
80. Amponsah L, Chuck C, Parsons S. Life cycle assessment of a marine biorefinery producing protein, bioactives and polymeric packaging material. Int J Life Cycle Assess. 2024;29(2):174–91.
81. Gaber K, Roesch C, Biondi N. Life cycle assessment of total fatty acid (TFA) production from microalgae *Nannochloropsis oceanica* at different sites and under different sustainability scenarios. Bioenerg Res. 2022;15(3):1595–615.
82. Zapata-Boada S, Gonzalez-Miquel M, Jobson M, Cuellar-Franca RM. Techno-economic and environmental analysis of algae biodiesel production via lipid extraction using alternative solvents. Ind Eng Chem. Res. 2022;61(49):18030–44.
83. Cai C, Chen X, Li F, Tan Z. Three-phase partitioning based on CO_2-responsive deep eutectic solvents for the green and sustainable extraction of lipid from *Nannochloropsis* sp. Sep Purif Technol. 2021;279:119685.
84. Motlagh SR, Harun R, Biak DRA, Hussain SA, Omar R, Elgharbawy AA. COSMO-RS based prediction for alpha-linolenic acid (ALA) extraction from microalgae biomass using room temperature ionic liquids (RTILs). Mar Drugs. 2020;18(2):108.
85. Fan C, Liu Y, Shan Y, Cao X. A priori design of new natural deep eutectic solvent for lutein recovery from microalgae. Food Chem. 2022;376:131930.
86. Li K, Jiang C, Han S-I, Kang S, Chen J, Won D, et al. Green and efficient method to acquire high-value phycobiliprotein from microalgal biomass involving deep eutectic solvent-based ultrasound-assisted extraction. Food Chem. 2024;449:139196.
87. Motlagh SR, Harun R, Biak DRA, Hussain SA, Wawak G, Khezri R, et al. Screening of suitable ionic liquids as green solvents for extraction of eicosapentaenoic acid (EPA) from microalgae biomass using COSMO-RS model. Molecules. 2019;24(4):713.
88. Xu D, Chow J, Weber CC, Packer MA, Baroutian S, Shahbaz K. Evaluation of deep eutectic solvents for the extraction of fucoxanthin from the alga *Tisochrysis lutea* - COSMO-RS screening and experimental validation. J Environ Chem Eng. 2022;10(5):108370.
89. Zapata-Boada S, Gonzalez-Miquel M, Jobson M, Cuellar-Franca RM. Integrating technoeconomic, environmental, and safety criteria in solvent screening for extraction processes: the case of algae lipid extraction. ACS Sust Chem Eng. 2022;10(1):472–85.
90. Lai H-L, Yang L-C, Lin P-T, Lai S-Y, Wang M-Y. Phagocytosis activity of three sulfated polysaccharides purified from a marine diatom cultured in a semi-continuous system. Int J Biol Macromol. 2020;155:951–60.

91. Koenig-Mattern L, Linke S, Rihko-Struckmann L, Sundmacher K. Computer-aided solvent screening for the fractionation of wet microalgae biomass. Green Chem. 2021;23(24):10014–29.
92. Koenig-Mattern L, Rihko-Struckmann L, Sundmacher K. Systematic solvent selection enables the fractionation of wet microalgal biomass. Sep. Purif Technol. 2025;354:129462.
93. Cosgun A, Gunay ME, Yildirim R. Machine learning for algal biofuels: a critical review and perspective for the future. Green Chem. 2023;25(9):3354–73.
94. Kavitha S, Ravi YK, Kumar G, Nandabalan YK, Banu JR. Microalgal biorefineries: advancement in machine learning tools for sustainable biofuel production and value-added products recovery. J Environ Manage. 2024;353.
95. Heisnam R, Keithellakpam OS, Kshetrimayum V, Mukherjee PK, Sharma N. Phycocyanin purified from *Westiellopsis* sp. induces caspase 3 mediated apoptosis in breast cancer cell line MDA-MB-231. Algal Res. 2022;68:102852.
96. Savio S, Turchi R, Carbone M, Ciavatta ML, Lettieri-Barbato D, Aquilano K, et al. Unveiling the anti-proliferative and pro-thermogenic activity of *Staurosirella pinnata* (Bacillariophyta) bioproducts. Algal Res. 2024;78:103393.
97. Derbel H, Elleuch J, Mahfoudh W, Michaud P, Fendri I, Abdelkafi S. In vitro antioxidant and anti-inflammatory activities of bioactive proteins and peptides from *Rhodomonas* sp. Appl Sci. 2023;13(5):3202.
98. Guzman F, Wong G, Roman T, Cardenas C, Alvarez C, Schmitt P, et al. Identification of antimicrobial peptides from the microalgae *Tetraselmis suecica* (Kylin) butcher and bactericidal activity improvement. Mar Drugs. 2019;17(8):453.
99. Lauritano C, Martin J, de la Cruz M, Reyes F, Romano G, Ianora A. First identification of marine diatoms with anti-tuberculosis activity. Sci Rep. 2018;8:2284.
100. Caires TA, da Silva AMS, Vasconcelos VM, Affe HMJ, de Souza Neta LC, Boness HVM, et al. Biotechnological potential of *Neolyngbya* (Cyanobacteria), a new marine benthic filamentous genus from Brazil. Algal Res. 2018;36:1–9.
101. Motlagh SR, Elgharbawy AA, Khezri R, Harun R, Omar R. Ionic liquid-based microwave-assisted extraction of protein from *Nannochloropsis* sp. biomass. Biomass Convers Bior. 2021;13:8327–38.
102. Bauer MC, Becker GC, Kruse A. Influence of water content and upstream processing on the liquefied dimethyl ether batch e*xtraction of Phaeodactylum tricornutum*. ACS Sustain Chem. Eng. 2022;10(49):16250–60.
103. Mahmood WMAW, Theodoropoulos C, Gonzalez-Miquel M. Enhanced microalgal lipid extraction using bio-based solvents for sustainable biofuel production. Green Chem. 2017;19(23):5723–33.
104. Olkiewicz M, Caporgno MP, Font J, Legrand J, Lepine O, Plechkova NV, et al. A novel recovery process for lipids from microalgae for biodiesel production using a hydrated phosphonium ionic liquid. Green Chem. 2015;17(5):2813–24.
105. Jimenez Callejon MJ, Robles Medina A, Gonzalez Moreno PA, Esteban Cerdan L, Orta Guillen S, Molina Grima E. Simultaneous extraction and fractionation of lipids from the microalga *Nannochloropsis* sp. for the production of EPA-rich polar lipid concentrates. J Appl Phycol. 2020;32(2):1117–28.
106. Zhang Y, Ward V, Dennis D, Plechkova NV, Armenta R, Rehmann L. Efficient extraction of a Docosahexaenoic acid (DHA)-rich lipid fraction from *Thraustochytrium* sp using ionic liquids. Materials. 2018;11(10):1986.
107. Zhang R, Wang J, Zhai X, Che J, Xiu Z, Chi Z. Carbonate assisted lipid extraction and biodiesel production from wet microalgal biomass and recycling waste carbonate for CO_2 supply in microalgae cultivation. Sci Total Environ. 2021;779:146445.
108. Sun J, Zhou C, Cheng P, Zhu J, Hou Y, Li Y, et al. A simple and efficient strategy for fucoxanthin extraction from the microalga *Phaeodactylum tricornutum*. Algal Res. 2022;61:102610.
109. Seeger J, Wendisch VF, Henke NA. Extraction and purification of highly active astaxanthin from *Corynebacterium glutamicum* fermentation broth. Mar Drugs. 2023;21(10):530.
110. Ruiz-Dominguez MC, Marticorena P, Sepulveda C, Salinas F, Cerezal P, Riquelme C. Effect of drying methods on lutein content and recovery by supercritical extraction from the microalga *Muriellopsis* sp. (MCH35) cultivated in the arid north of Chile. Mar Drugs. 2020;18(11):528.
111. Ruiz-Dominguez MC, Salinas F, Medina E, Rincon B, Ma M, Gutierrez MC, et al. Supercritical fluid extraction of fucoxanthin from the diatom *Phaeodactylum tricornutum* and biogas production through anaerobic digestion. Mar Drugs. 2022;20(2):127.
112. Paterson S, Villanueva-Bermejo D, Hernandez-Ledesma B, Gomez-Cortes P, de la Fuente MA. Supercritical CO_2 extraction increases the recovery levels of omega-3 fatty acids in *Tetraselmis chuii* extracts. Food Chem. 2024;453:139692.
113. Zinnai A, Sanmartin C, Taglieri I, Andrich G, Venturi F. Supercritical fluid extraction from microalgae with high content of LC-PUFAs. A case of study: Sc-CO_2 oil extraction from *Schizochytrium* sp. J Supercrit Fluid. 2016;116:126–31.

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